

Operational Efficiency of Photovoltaic (PV) Modules

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Abstract— As solar energy becomes more vital, maximizing the performance of photovoltaic (PV) modules is essential for ensuring reliable and sustainable energy generation. However, several environmental and manufacturing factors can greatly reduce the efficiency of PV panels, making accurately assessing and categorizing the impact of these factors on energy production crucial for effective maintenance, diagnostics, and optimization of solar installations. For that purpose, this project employed several machine learning and deep learning models, evaluating and comparing their performances to understand their strengths and drawbacks.

The analysis of the data indicated a continuous variance of features' values across train, validation and test sets, with the numeric label following a Gaussian distribution and a class imbalance for the categorical label. The PCA revealed that the most relevant features of the dataset were 'affected_area' and 'ISC'.

For both regression and classification tasks, the top-performing model was the Feed Forward Neural Network, with MSE and R^2 of 0,0350 and 1,0000 for 10 and 2 features, and for classification, accuracies of 99,63% and 76,51% for 10 and 2 features. The models evidenced a clear overfitting to the data due to the homogeneous variance on the different sets, making it impossible to draw conclusions regarding their generalization capability.

Index Terms— Photovoltaic Modules, Regression, Multi-class Classification, Machine Learning, Deep Learning

I. INTRODUCTION

Automated decisions are driving the forefront of technological advancements, revolutionizing industries by empowering systems to execute tasks with minimal human intervention. In the context of renewable energy management, automated approaches play a pivotal role in optimization energy production, distribution and maintenance [1]. Predictive models estimate energy efficiency, providing accurate forecasts of energy output and identifying inefficiencies (e.g caused environmental or manufacturing factors), that may affect the performance of PV modules.

This project envisioned the implementation of models to predict two outputs: expected efficiency (continuous-valued variable) and efficiency level (categorical variable) for a given dataset, regarding PVs' features and its annotations (targets). Therefore, the goal was to develop predictive models that can perform both tasks, ensuring robustness and ability to generalize efficiently to different contexts. To address this

topic, we propose three data-driven fully supervised learning (FSL) models: Random Forest (RF), Support Vector Machine (SVM), combined with Linear Regression, and Feed Forward Neural Network (FFNN), to comply with both classification (efficiency level) and regression (expected efficiency) tasks.

RF is an ensemble learning method that builds multiple decision trees during training and aggregates their predictions to make a final decision. In classification, it outputs the most frequent class (mode) from the individual trees, while in regression, it calculates the average prediction of all trees. Ensemble methods combine several models to enhance overall performance by reducing variance, bias, or both. RF introduces randomness into the learning process in two key ways: (i) by using bootstrapped subsets of the training data (a technique known as bagging), and (ii) by randomly selecting a subset of features for splitting at each node. This randomness helps mitigate overfitting and reduces variance, making the model more robust and reliable [2].

SVMs are a set of SL methods used for classification, regression and outliers' detection. SVM constructs a hyperplane or set of hyper-planes in a high or infinite dimensional space, which can be used for classification, regression or other tasks. Intuitively, a good separation is achieved by the hyperplane that has the largest distance to the nearest training data points of any class (so-called functional margin), since in general the larger the margin the lower the generalization error of the classifier [3].

Linear regression is a supervised learning algorithm used to predict continuous values by finding a straight line or hyperplane that best fits a set of data points. It establishes a linear relationship between the input variables (features) and the output variable (target). The goal is to determine the line's parameters that minimize the differences (squared errors) between predicted and actual values. Its simplicity and effectiveness make linear regression a core technique in Machine Learning (ML) [4].

Feed Forward Neural Networks (FFNNs) are a type of Dense Neural Network (DNN), meaning they are fully connected, with information flowing in one direction (e.g input to output layer, through hidden layers).

FFNNs excel in learning complex patterns capabilities, capturing intricate relationships between inputs and outputs. Increasing the number of hidden layers can grow the NN's capability to learn more abstract features and representations. However, higher-order interactions introduce challenges as overfitting and vanishing gradients (i.e. diminishing of gradi-

ents during backpropagation), compromising its capability to generalize or adapt to unseen data. To mitigate these issues, it is crucial to employ regularization techniques (e.g., dropout, L2 regularization) and perform hyperparameter tuning (e.g., adjusting the number of hidden layers) to optimize the network's architecture and improve its performance.

The proposed methods underwent a training and a testing phase, and their performances were compared in terms of evaluation metrics for classification and regression tasks.

II. METHODOLOGY

For the purpose of this work, the chosen dataset, designed to support models that estimate the efficiency of photovoltaic (PV) modules, accounts for 100.000 observations of several key characteristics and defect indicators that affect the modules' performance.

The dataset consists of 10 input features, with 5 of them categorical and the other 5 numerical, and two types of labels, a continuous variable for regression and a multiclass label for classification.

TABLE I
DATASET'S FEATURES AND LABELS AND THEIR DESCRIPTION

Features	Description
pv_module_type	Monocrystalline, Polycrystalline, Thin-Film
hotspot	Severe Hotspot, Mild Hotspot, None
birddrop	Many, Few, None
soiling	High, Moderate, Low, None
junction_box	Visible, None
affected_area	Percentage of module's area affected by defects
temperature	Average temperature of the PV module
irradiance	Irradiance level of the PV module
Voc	Voltage in open circuit
Isc	Current in short circuit
Labels	Description
expected_efficiency	Percentage estimate of the PV module's efficiency
efficiency_level	good, moderate, bad, extremely_bad

A. Data Processing

1) *Dataset Statistical Analysis*: To statistically evaluate the dataset's numerical features, the mean and standard deviation of each were computed. As for the categorical features, since it is not possible to compute those measurements, the chosen metric was frequency distribution, which indicates the number of observations for each category.

In terms of labels, to assess the distribution of the continuous label, the samples were reorganized in an histogram representation with 30 bins. As for the categorical label, since this dataset is suitable for multiclass classification, this variable was evaluated according to its class distribution, representing the number of occurrences of a class according to the size of the dataset.

To understand the characteristics of the training, validation, and test sets used throughout this work, the aforementioned metrics were computed for each set individually, offering valuable insights into each phase.

2) *Data Processing*: The data processing stage plays a crucial role in ensuring that the dataset is well-suited for training machine learning models, as improper handling of data can significantly impact a model's training performance and generalization capability. To prepare the dataset for regression and classification tasks, a comprehensive processing pipeline was developed, addressing both numerical and categorical features as well as the target variables.

For categorical features, such as *pv_module_type* and *hotspot*, one-hot encoding was applied. This technique transforms categorical values into binary indicator variables, allowing them to be effectively incorporated into machine learning algorithms. The encoding process was performed using the *OneHotEncoder* from the *scikit-learn* library, which was first fitted on the training data and then consistently applied to the validation and test subsets to maintain uniformity across all splits.

Numerical features, including *affected_area*, *temperature*, and *irradiance*, underwent a two-step scaling procedure to standardize their ranges. First, the features were normalized to a range of 0 to 1 using *MinMaxScaler*, ensuring compatibility with algorithms sensitive to feature scales. Subsequently, these normalized values were standardized to have zero mean and unit variance using *StandardScaler*, facilitating effective training by mitigating the influence of differing feature magnitudes.

The processed numerical and categorical features were combined to form a single dataset for each subset (training, validation, and test). This integration ensured that the processed features were properly aligned and ready for model input. The unified feature sets maintained the original dataset's structure while enhancing its compatibility with machine learning workflows.

The target variable, *efficiency_level*, a multiclass categorical label, was mapped to numerical values: 0 for *extremely_bad*, 1 for *bad*, 2 for *moderate*, and 3 for *good*. This mapping was required for some of the methods explored.

This robust approach to data preparation resulted in a dataset that was well-structured and suitable for model development and evaluation across regression and classification tasks.

3) *Principal Component Analysis (PCA)*: PCA is a statistical technique employed for dimensionality reduction. This method proposes to find the dataset's principal components, which represent the directions of maximum variance in the data, i.e., the directions that capture the greatest spread [5].

In this project, PCA was implemented to identify the two most important dataset features, based on their contribution to the principal components. These features were then fed into the distinct models, and the remaining ones were discarded. This approach was intended to assess the influence of simplifying the model (e.g., reducing the number of parameters) on its performance, potentially helping it to generalize better to unseen data. Moreover, by focusing on the most important features, the model can filter out noisy data and uncover the underlying relationships within the dataset, leading to reliable predictions.

B. Models' Implementation

1) *Linear Regression*: The Linear Regression Model uses the Ordinary Least Squares method, which estimates the model's parameters by minimizing the sum of the squared errors [6], the cost function defined in Eq. (1):

$$J(\theta_0, \theta_1, \dots) = \sum_{n=1}^N (y_n - \theta_n x_n)^2 \quad (1)$$

The model was trained with the concatenation of the train and validation sets and tested on the test set in two separate occasions. It was first trained with all the 10 numeric and categorical features of the dataset, and then trained with only the most relevant features according to the PCA.

2) *Random Forest (RF)*: The Random Forest Model was performed with both classification and regression objectives. Firstly, all 10 numerical and categorical features provided in the dataset were utilized, secondly, a selection was employed to retain only the most relevant features identified via PCA. The training process included concatenating the training and validation sets into a single training-validation data set, using a predefined split for cross-validation. Hyperparameter tuning was performed through grid search with the following parameter ranges:

TABLE II
HYPERPARAMETERS' TUNNING

Hyperparameter	Search Space
Number of trees in the forest	50, 100, 200
Maximum depth of each tree	10, 20, None
Minimum samples required to split a node	2, 5, 10

The best-performing model was selected based on its accuracy, in the classification task, and on the Mean Squared Error (MSE), in the regression task.

For classification, the RF model assigns a class label based on the majority vote of the trees, defined aiming to minimize the Gini Impurity at each split, defined as (2):

$$G = \sum_{k=1}^K p_k(1 - p_k), \quad (2)$$

where p_k is the proportion of samples belonging to class k at a node.

For regression, the RF model predicts a continuous value as the average of all individual tree predictions. Uniform sampling was inherently performed during the model's default bootstrap aggregation process, ensuring equal likelihood of selection for all data points and the cost function used for regression minimizes the Mean Squared Error (MSE), as defined in: (3):

$$J(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (3)$$

where y_i and $\hat{y}_i$ are the true and predicted values, respectively, and N is the number of samples.

3) *Support Vector Machine (SVM)*: The first step to define the model is to determine the best parameter configuration, called hyperparameter tuning. For that purpose, the grid search method was employed. This method systematically evaluates all possible combinations from a predefined parameter grid by fitting the model on the available data samples and comparing the resulting accuracy. The grid search was performed with a predefined split cross validation, where the train was performed on the train set samples and the test on the validation set.

TABLE III
HYPERPARAMETERS' TUNNING

Hyperparameter	Search Space
Kernel	'linear', 'rbf'
C	0.1, 1, 10
gamma	'scale', 'auto'

The 'kernel' parameter defines the type of decision boundary the model will use, and in this case we test a linear kernel and a Radial Basis Function kernel. The 'C' parameter is known as the regularization parameter and controls the trade-off between achieving a low training error and a low testing error, penalizing the model for misclassification. 'gamma' is the kernel's coefficient for the rbf kernel and controls how far the influence of a single training example reaches. In addition, the parameter 'class_weight' assigns weights to the classes, helping when dealing with imbalanced datasets. The weight is inversely proportional to class frequencies in the training set if set to 'balanced' [7].

The loss function implemented by this method is described by Eq. (4), and our goal is to find $w \in \mathbb{R}^p$ and $b \in \mathbb{R}$ such that the prediction given by $\text{sign}(w^T \phi(x) + b)$ is correct for most samples.

$$\begin{aligned} \min_{w, b, \zeta} \quad & \frac{1}{2} w^T w + C \sum_{i=1}^n \zeta_i \\ \text{subject to} \quad & y_i(w^T \phi(x_i) + b) \geq 1 - \zeta_i, \\ & \zeta_i \geq 0, \quad i = 1, \dots, n \end{aligned} \quad (4)$$

Has this is a multi-class classification problem, the model employs the One-vs-Rest, by splitting the multi-class dataset into multiple binary classification problems. A binary classifier is then trained on each binary classification problem and predictions are made using the model that is the most confident.

Having established the best parameter configuration, the model was evaluated on unseen test set samples using appropriate evaluation metrics. This procedure was performed both using all the features of the dataset and only the most important features identified through PCA.

4) *Deep Learning, Feed Forward Neural Network (FFNN)*: Regarding the model architecture, a multitask NN was implemented to perform both regression and classification assignments. This framework leverages a ReLU-activated shared hidden layer to learn a common representation for both tasks, while maintaining two parallel task-specific output layers, Fig. (1).

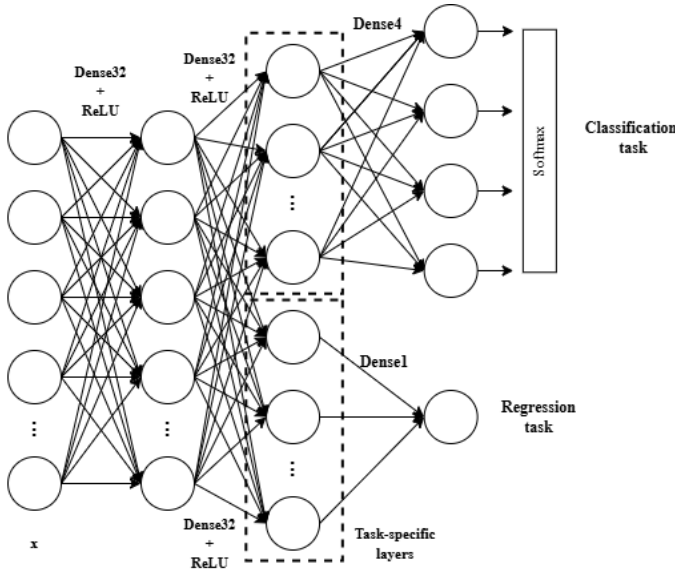


Fig. 1. Multi-task Feed Forward Neural Network (Multi-task FFNN)

The model handles regression with a single-node linear output layer without an activation function, while classification is managed through a softmax-activated output layer. Accordingly, the loss function was defined as a weighted sum of regression and classification losses, as shown in Eq. (1):

$$\text{total_loss} = \alpha \times \text{regression_loss} + \beta \times \text{classification_loss} \quad (5)$$

where the regression_loss and classification_loss are computed using MSE and cross-entropy, Eq.(6), respectively.

$$\text{cross-entropy} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C y_{ij} \log(\hat{y}_{ij}) \quad (6)$$

α and β control the relative importance of each task.

In addition, a dropout layer was applied to the hidden layer to attempt to mitigate overfitting. Dropout is a regularization technique in which a random subset of neurons, determined by the dropout rate, is temporarily “dropped” (i.e set to zero) in each iteration. This prevents the model from becoming overly reliant on specific neurons, encouraging it to learn more robust patterns, rather than idiosyncratic features [8]. The dropout rate was set to 0.3, meaning 30% of the neurons in the second layer were deactivated in each forward pass.

The model training process employed gradient descent optimization, using Adam optimizer to update the network parameters. Adam optimizer combines momentum and adaptive learning rates (for each parameter), making it highly efficient for training deep NNs. It handles noisy data by reducing the gradients’ estimates variance, leading to more stable and consistent updates. As to the training process, the Adam optimizer was used with a specific learning rate (lr) and weight decay (wd), which supports the penalization of large weights in L2 regularization. To complement this framework, a learning rate scheduler was incorporated to dynamically adjust lr during training, facilitating a more effective optimization process and enhancing loss convergence.

The model was initially trained using all available input features, enabling the NN to learn from the complete set of

information. Additionally, training was also conducted with the two most important features, identified through PCA. Validation was performed throughout both training phases to provide an estimate of the model’s performance on unseen data.

C. Evaluation Metrics

To evaluate the regression task, the Mean Squared Error (MSE) and R^2 metrics were computed. The MSE measures the average squared difference between the predicted values and the true values, providing a way to assess the predictor’s accuracy, with lower values indicating better performance. The R^2 metric, also known as the coefficient of determination, quantifies the proportion of variance in the dependent variable that is explained by the independent variables. In other words, R^2 evaluates the model’s goodness of fit, indicating how well the predictions align with the actual data.

As for the classification task, several metrics were employed to precisely understand the models’ performance.

TABLE IV
EVALUATION METRICS DESCRIPTION

Evaluation Metric	Description
Accuracy	Proportion of correct predictions.
Balanced Accuracy	Average recall across all classes.
AUC-ROC Score	Area under the ROC curve.
Precision	Proportion of true positives among all positives.
F1-score	Harmonic mean of precision and recall.
Recall	Proportion of actual positives correctly predicted.

Since this is a multi-class classification problem, the AUC-ROC Score, Precision, Recall and F1-score required averaging, done by weighting according to the classes’ distribution.

To further evaluate the accuracy of a classification and understand the performance of the model for each class, the confusion matrix is computed. By definition a confusion matrix C is such that is $C_{i,j}$ equal to the number of observations known to be in group i and predicted to be in group j .

III. RESULTS AND DISCUSSION

A. Data Processing

1) *Dataset Statistical Analysis*: The results for the statistical analysis of the dataset’s features are introduced by Table VII in the appendix. It shows that across all sets, the values of mean, standard deviation and relative frequency are maintained with slight deviations, ensuring that the differences of the performances of the models aren’t due to different distributions of the observations among the separate sets.

The information presented in Fig. (2) indicates that the observations in the train, validation, and test sets follow the same distribution, both for the numeric label, the expected efficiency, and for the categorical label, the efficiency level.

For the expected efficiency, the data follows a Gaussian distribution with similar mean and standard deviation rounding 0.188% and 0.033. As for the efficiency level, the distribution of observations among classes is also homogeneous and equal to 0.512 for ‘good’, 0.265 for ‘extremely_bad’, 0.158 for ‘bad’ and 0.064 for ‘moderate’. It is obvious the class imbalance

in this dataset and in the individual sets, where the class 'good' has the most amount of observations and 'moderate' the lowest.

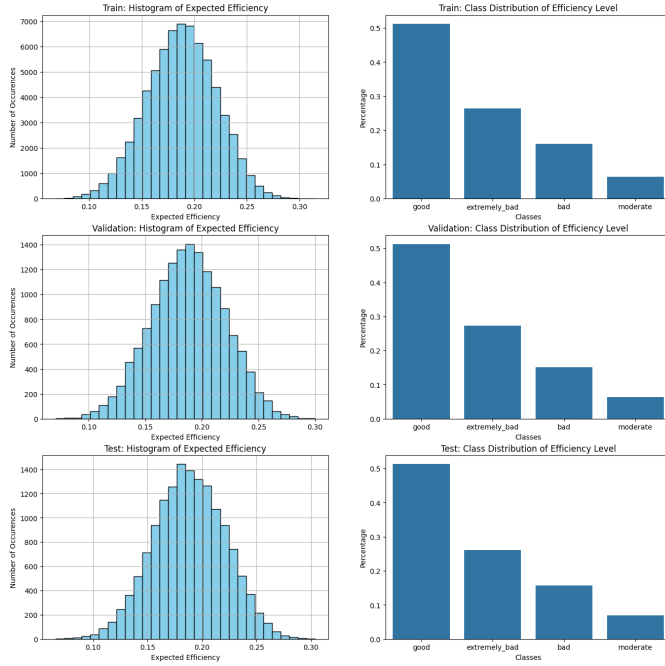


Fig. 2. Expected efficiency and efficiency level labels' distributions on train, validation and test sets.

2) *Principal Component Analysis (PCA)*: PCA identified Short-Circuit Current, ISC, and relative affected area of the PV modules as the two most important features in the dataset, contributing 46% and 25% to the principal components, respectively. These features exhibit a strong negative correlation, with ISC decreasing as the affected area increases, Fig.(3).

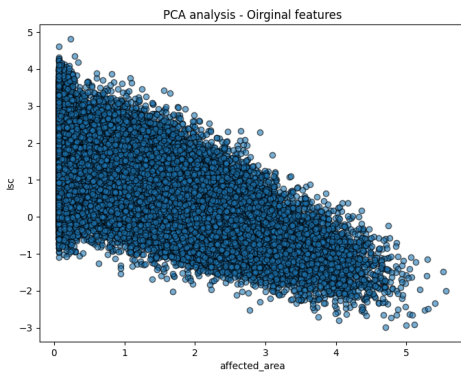


Fig. 3. Visual representation of the relationship between ISC and relative affected area

ISC provides insight into the overall electrical capacity of the solar system, as it is a key indicator of the panel's ability to generate current. Relative affected area of the module's surface could lead to reduced efficiency, as its damaged area cannot effectively absorb sunlight or generate electricity [9].

TABLE V
RESULTS OBTAINED FOR CLASSIFICATION TASK

Methods	Acc	Balanced Acc	ROC-AUC	F1	Recall	Precision
RF, 10 features	0,9744	0,9358	0,9992	0,9739	0,9744	0,9743
SVM, 10 features	0,9883	0,9896	1,0000	0,9886	0,9883	0,9894
FFNN, 10 features	0,9963	0,9925	1,0000	0,9963	0,9963	0,9964
RF, 2 features	0,6949	0,5775	0,9103	0,7017	0,6949	0,7097
SVM, 2 features	0,6595	0,6981	0,9341	0,6938	0,6595	0,7988
FFNN, 2 features	0,7651	0,5535	0,9376	0,7322	0,7651	0,7342

B. Models' Implementation

For the classification task using all available features, the method that most accurately predicted the expected efficiency level was the FFNN, achieving an impressive accuracy of 99.63% on the test set. In addition, both RF and SVM delivered strong performance, with accuracy rates of 97.44% and 98.83%, respectively.

When assessing balanced accuracy, RF showed greater difficulty in managing the inherent class imbalance within the dataset, achieving a lower value, 93,58%, when compared to SVM and FFNN. Additionally, SVM and FFNN achieved perfect AUC scores, highlighting their exceptional performance across all classification thresholds. Lastly, the F1 score, which balances precision and recall, is particularly relevant in imbalanced classification tasks like this one, as it reflects the models' ability to minimize both false positives and false negatives. The consistently high F1 scores across all methods indicate their reliability for this task, with the FFNN standing out as the top performer.

Regarding the results obtained using just the 2 most significant features for classification, the overall performance of all methods decreased significantly. While FFNN keeps being the method with the highest accuracy, 76,51%, RF overcomes SVM in the proportion of correct predictions. The opposite occurs in terms of balanced accuracy, indicating that the SVM model, with a balanced accuracy of 69,81%, handles class imbalance better than RF and FFNN. Furthermore, the value for balanced accuracy overcomes the accuracy, indicating that SVM has a better prediction for the minority classes. As for the metrics of ROC-AUC score, F1 score and Recall, FFNN remains the highest performing model, although it falls behind SVM in terms of Precision, with a value of 0,7988, which suggests that SVM is more reliable when classifying positive cases, ensuring a higher proportion of its positive predictions are correct.

The significant drop in performance across all metrics and for all models when reducing the number of features from 10 to 2 highlights the importance of a more complete feature set to capture the complexities of the classification problem.

Overall, FFNN demonstrates superior performance across most metrics, particularly in terms of accuracy and recall, regardless of the number of features used. However, the SVM model shows stronger precision, especially with fewer features, making it a viable choice for applications prioritizing reliability in positive classifications.

Regarding regression performance across all available features, RF achieved a null MSE and high R^2 , 0.9999, explaining nearly all data variance. In contrast, FFNN exhibited a significantly higher MSE, 0.0349, compared to ML methods,

TABLE VI
RESULTS OBTAINED FOR REGRESSION TASK

Methods	MSE	R ²
RF, 10 features	0.0000	0.9999
Linear Regression, 10 features	1,4115x10 ⁻⁶	0.9987
FFNN, 10 features	0.0349	1.0000
RF, 2 features	0.0001	0.9446
Linear Regression, 2 features	5,9045x10 ⁻⁵	0.9450
FFNN, 2 features	0.0351	1.0000

while maintaining a perfect R². This discrepancy suggests that, although FFNN effectively captures the general trends in the data, it lacks in individual precision, likely due to the smoothness effect imposed by activation functions (ReLU). Linear Regression also demonstrated strong performance but fell slightly short of RF's precision, as it is less adept at capturing more intricate non-linear relationships.

Based on these results, RF stands out as the most effective method, offering robustness and the ability to model complex patterns proficiently.

Analyzing the results for the two most important features, it is observed that both RF and Linear Regression models exhibit a significant decline in performance, with their R² values dropping to 0.9446 and 0.9450, respectively. This implies that feature reduction has a detrimental impact on these algorithms, as they appear to be more sensitive to the removal of key features. In contrast, FFNN remains unaffected by the reduction of categorical features, indicating that the regression task in FFNNs is likely more dependent on numerical features that sustain predictive performance.

Therefore, in a reduced categorical feature situation, choosing the FFNN model would be beneficial, as it remains robust and effectively captures complex interactions.

Ultimately, FFNN appears to outperform the ML methods for both classification and regression. However, since the testset shares the same variance as the training set, it is not possible to conclusively determine the model's ability to generalize effectively to unseen data. This raises the possibility that the model's performance might result from overfitting due to the memorization of the training data, rather than effectively capturing the underlying patterns in the data.

IV. CONCLUSION

This report suggests different ML and DL approaches to the task of assessing the impact of environmental and manufacturing factors on a photovoltaic module's efficiency in energy production in a period marked by the growing importance of solar energy. The models were evaluated with all the features of the dataset and the most relevant according to the PCA's results, which are 'affected_area' and 'Isc'.

A statistical analysis of the train, validation and test sets indicates a clear imbalance of the efficiency level classes, which might affect on the models' performance, being reflected in the balanced accuracy metric, as it corrects for imbalances by giving each class equal weight. Furthermore, the homogeneous variance of the features and distribution of the expected efficiency across all sets, paired with the

exceptional results when 10 features are used suggests a degree of overfitting across all methods.

For the regression task, the best performing approach was the FFNN, for 10 and 2 features. The fact that there was no reduction in the value of R² suggests that this model is more flexible than the other even with less data. Its relatively high MSE value evidence overfitting to the data.

Regarding classification, the decrease on input features takes a toll on models' performance, reinforcing the importance of the relationships between features that influence the classification task. While FFNN maintains its position as the top-performing model for 10 features, there is a slight competition with SVM with 2 features, as that method achieves higher values for balanced accuracy and precision.

To improve the models' performance, a wider search for a combination of parameters for RF and SVM models could be employed. Incorporating other machine learning and deep learning algorithms, such as Lasso and Ridge Regression, a k-NN algorithm or convolutional neural networks, could further enrich the evaluation process by providing a diverse set of models to compare performance across different approaches.

Since overfitting is a clear issue, implementing approaches such as data augmentation and out-of-distribution testing to modify the test set variance could overcome that problem. Additionally, implementing L1/L2 regularization for some approaches and out-of-domain testing would provide valuable insights into the generalization capability of the presented models, as these tests involve data that differs significantly from the training set and may reveal how well the models handle unfamiliar or atypical scenarios.

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V. APPENDIX

TABLE VII

SUMMARY STATISTICS AND RELATIVE FREQUENCIES FOR TRAIN, VALIDATION, AND TEST SETS

Metric	Train	Validation	Test
Mean $\pm$ STD			
Affected Area	0.12 ± 0.13	0.12 ± 0.13	0.12 ± 0.13
Temperature	24.97 ± 4.98	25.07 ± 5.05	24.98 ± 5.03
Irradiance	999.77 ± 99.69	1000.10 ± 100.08	999.76 ± 99.62
Voc	44.90 ± 1.07	44.90 ± 1.06	44.92 ± 1.08
Isc	10.00 ± 1.57	10.01 ± 1.58	10.02 ± 1.57
Relative Frequency			
PV Module Type			
Poly-BSF	0.25	0.25	0.25
HJT	0.25	0.25	0.26
Thin-Film	0.25	0.24	0.25
Mono-PERC	0.25	0.25	0.25
Hotspot			
None	0.47	0.46	0.47
Mild Hotspot	0.27	0.27	0.26
Severe Hotspot	0.26	0.27	0.27
Birddrop			
None	0.47	0.47	0.47
Few	0.27	0.27	0.27
Many	0.27	0.27	0.27
Soiling			
None	0.40	0.39	0.40
Low	0.20	0.20	0.20
Moderate	0.20	0.20	0.20
High	0.20	0.20	0.20
Junction Box			
None	0.60	0.60	0.60
Visible	0.40	0.40	0.40

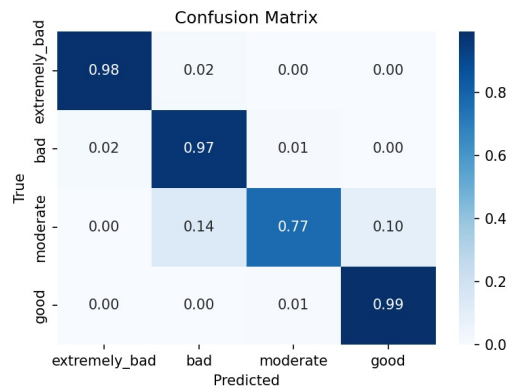


Fig. 4. RF Confusion Matrix with 10 features.

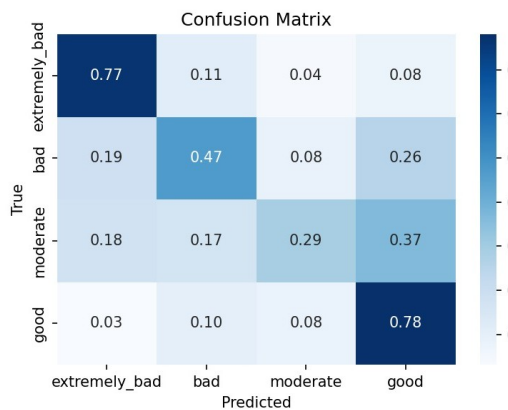


Fig. 5. RF Confusion Matrix with 2 features.

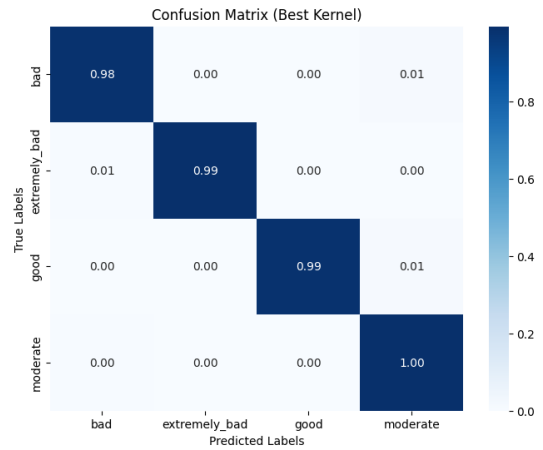


Fig. 6. SVM Confusion Matrix with 10 features.

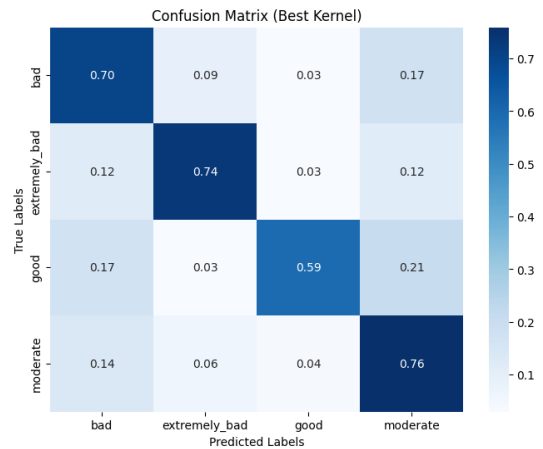


Fig. 7. SVM Confusion Matrix with 2 features.

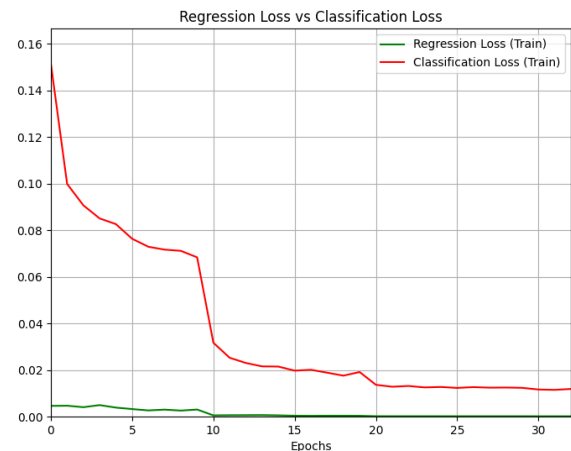


Fig. 8. FFNN Regression and Classification Losses with 10 features.

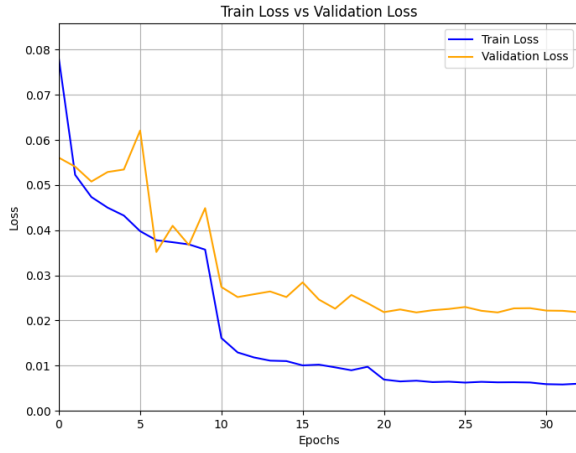


Fig. 9. Train and Validation Loss of FFNN with 10 features.

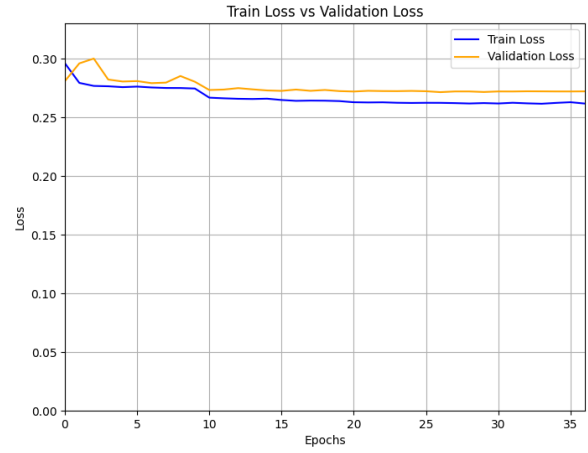


Fig. 12. Train and Validation Loss of FFNN with 2 features.

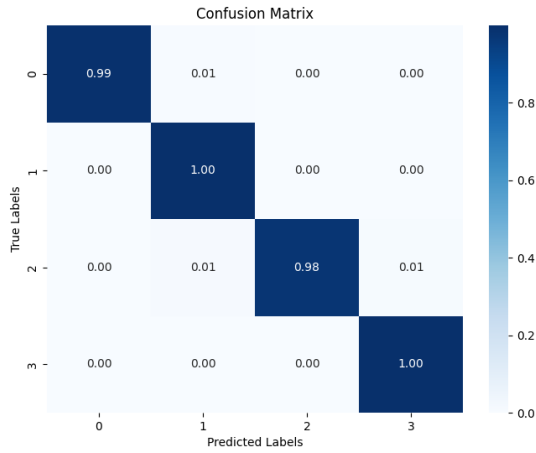


Fig. 10. FFNN Confusion Matrix with 10 features.

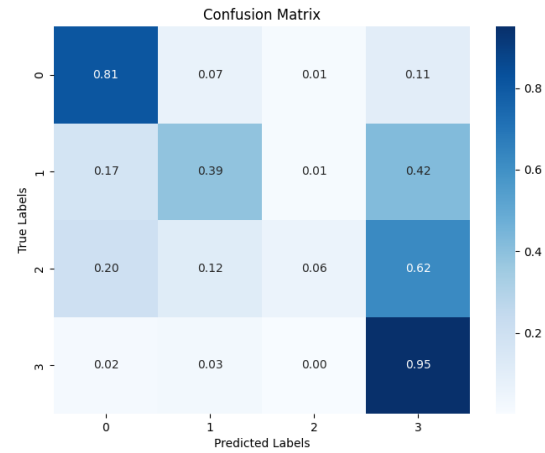


Fig. 13. FFNN Confusion Matrix with 2 features.

VI. CONTRIBUTIONS

- Filipa Pato:** RF and Data Preprocessing Software; RF and Data Preprocessing Methodology, Classification Results - 33.3%
- Leandro Ribeiro:** LR, SVM, Statistical Analysis and Data Preprocessing Software; LR, SVM, Statistical Analysis and Data Preprocessing methodology, Statistical Analysis Results, Abstract, Conclusion - 33.3%
- Leonor Franco:** FNN and PCA Software, FNN and PCA Methodology, Introduction, Regression and PCA Results - 33.3%

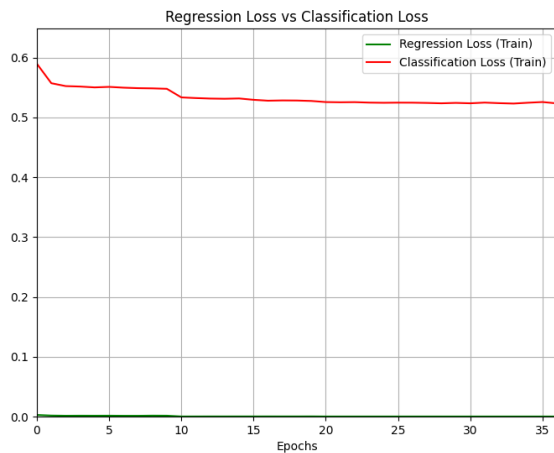


Fig. 11. FFNN Regression and Classification Losses with 2 features.